

SIMONA (SIMIN) LIU

647-677-4158 • simona.liu@mail.utoronto.ca • [linkedin.com/in/simonalIU](https://www.linkedin.com/in/simonalIU) • github.com/Ximonaaaaa

EDUCATION

BASc., Computer Engineering + PEY *University of Toronto*

Graduating May 2027 (Expected)

Minor in Business, Minor in Artificial Intelligence

Related Courses: Machine Learning, Deep Learning, Operating Systems, Comp. Algorithm & Data Structures, Control System, Computer Organization, Software Design, Accounting & Finance

TECHNICAL SKILLS

Project Management: Git, Jira, Wiki Page, Confluence Page, Excel, LaTeX

Programming: C++, C, PHP, Python, JavaScript, MySQL, Boomi, CSS, MATLAB, Verilog, Linux Shell Scripting

Certifications: Introduction to Generative AI for Software Development by DeepLearning.AI

PROFESSIONAL EXPERIENCE

Software Engineer Intern | *PHP (Laravel), AWS, MySQL*

July 2025 – Present

Uni Express Inc. • Richmond, BC

- Developed and optimized backend features for UniUni's internal website, UniSettlement, **improving the automation** of billing processes and data extraction.
- Initiated and designed backend structures for UniSettlement's Merchant Management project, allowing Customer Success team to record and edit merchants' information in database, reducing 60% of administrative efforts.
- Built SuiteScripts and maintained Boomi logics to integrate UniSettlement with NetSuite, ensuring over hundreds of billing files to be synchronized simultaneously between systems.

RESEARCH

Lead Researcher

March 2025 – Oct 2025

Betting on Dual-Use Technology: How AI and Marketing Rewires Modern Gambling

- Conducted a comprehensive literature review on technology and marketing trends in the wagering industry.
- Synthesized findings on personalization methods, user engagement models, and key ethical risks associated with AI-enabled betting platforms.

PROJECTS

Graphical User Interface System (GIS) | *C++, CSS*

Jan 2024 – Apr 2024

Led a team to develop a user-friendly GIS using the OpenStreetMap database and focusing on optimization.

- Designed pop-out window on user click with easily readable conventions and a responsive time of **0.3 seconds**.
- Scored **top 10** in developing an algorithm for a Travelling Salesman Problem (TSP) with geographic data of different cities using A*, Dijkstra and ant colony optimization.

Bank Card Recognition | *Python*

Aug 2024 - Dec 2024

Developed a deep learning model for recognizing the card number from a photo of a bank card using **YOLO & CRNN**.

- Developed baseline model using SVM classifier and HOG feature extractor, achieving a **60+%** accuracy.
- Preprocessed data through greyscaling and normalization, and augmented data through random rotations, scaling and cropping to increase data capacity from **700 to 3000** images; improved test accuracy from **65% to 86.4%**.
- Used the Ultralytics YOLOv5 architecture to implement our YOLO model consisting of **157 layers and 7,012,822 parameters** to achieve a precision of **over 99%**.

Embedded OS with Minesweeper | *C, NIOS II*

Jan 2024 - Apr 2024

Built Minesweeper as a **multi-threaded application** running on a NIOS II soft-core processor.

- Developed a Minesweeper game with randomized mine placement, grid-based input processing, flagging, uncovering of tiles and win/loss state detection with a restart system for continuous playability.
- Designed a **memory management system** for process isolation that allocates and frees memory dynamically without interfering with each other, allowing multiple application windows to run concurrently.